

TC2453

Machine Learning

Machine Learning Group Project Assignment

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1.0 Introduction

Credit card fraud is a major global financial concern, causing billions in losses annually, and financial institutions greatly benefit from early detection systems. In this project, our team will use deep learning (DL) and machine learning (ML) techniques to address the Credit Card Fraud Detection problem. The objective is to create a predictive model that, using a tabular dataset of transaction features, can reliably identify fraudulent transactions and protect customers from financial losses.

The following requirements for this project are in line with the course learning outcomes (CLOs). First, to use appropriate ML/DL techniques to train models, engineer features, and preprocess data and to identify the best method for fraud detection, analyze and compare model performance using relevant metrics.

We will start with exploratory data analysis (EDA) to address the severe class imbalance in the dataset (only 0.17% fraud cases), handle missing values, and discover transaction patterns. We will then preprocess the data and engineer features specifically designed for CNN architecture, including cyclical time encoding and spatial data transformation. Finally, we will train and evaluate multiple ML/DL models, including Random Forest, Logistic Regression, and our innovative Convolutional Neural Network approach that adapts computer vision techniques for tabular fraud data.

Our goal is to provide a robust predictive system by the project's conclusion, which will help financial institutions detect fraud in real-time and improve customer protection. The CNN approach represents a novel application of deep learning to financial fraud detection, demonstrating how computer vision techniques can be creatively adapted for tabular data analysis.

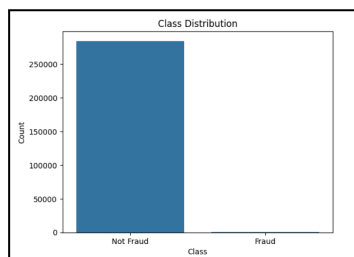
2.0 Exploratory Data Analysis (EDA) and Preprocessing

2.1 Dataset Overview

This project uses the **Credit Card Fraud Detection** dataset from Kaggle. It consists of transactions made by European cardholders in September 2013. The dataset contains **284,807 transactions**, with **492 labeled as fraud**. The features include **Time**, **Amount**, and **28 anonymized features (V1 to V28)** generated through a PCA transformation due to privacy reasons. The target variable is **Class**, where **0** represents non-fraudulent transactions and **1** indicates fraudulent transactions.

2.2 Class Distribution

To assess the balance between classes, we analyzed the distribution of the **Class** feature. We found a **highly imbalanced dataset** where only around **0.17%** of the transactions are labeled as fraud. This imbalance is visually represented in the bar plot and confirmed by the value counts. Such imbalance can negatively impact model performance if not handled, leading to bias toward the majority class. In the modeling phase, we plan to explore techniques like **SMOTE**, **undersampling**, or **class weighting** to address this issue.



```
Class counts and percentage:
Class
0    284315
1         492
Name: count, dtype: int64

Class
0    99.83
1     0.17
Name: count, dtype: float64
```

Diagram 1.0 Class Distribution

2.3 Missing Value Check

A missing value check was performed across all columns. The result showed that the dataset contains **no missing values**, so no imputation or data cleaning was necessary at this stage.

```
Missing values in each column:
Time      0
V1        0
V2        0
V3        0
V4        0
V5        0
V6        0
V7        0
V8        0
V9        0
V10       0
V11       0
V12       0
V13       0
V14       0
V15       0
V16       0
V17       0
V18       0
V19       0
V20       0
V21       0
V22       0
V23       0
V24       0
V25       0
V26       0
V27       0
V28       0
Amount    0
Class     0
dtype: int64
```

Diagram 2.0 Missing Value Check

2.4 Outlier Detection (Amount)

We used boxplots to inspect the **Amount** and **Time** features for outliers. The plot for **Amount** revealed several transactions with unusually high values, which could potentially skew the distribution and affect model performance. While these high-value transactions may indicate fraud, they can also be legitimate. Therefore, rather than removing them, we applied feature scaling to normalize the values. This ensures that algorithms sensitive to feature magnitude—such as logistic regression and support vector machines (SVMs)—can perform effectively without being biased by large numerical ranges.

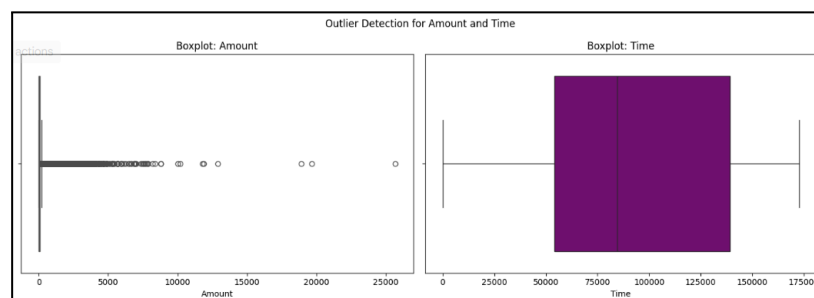


Diagram 3.0 Boxplot for Amount and Time

2.5 Feature Scaling

To standardize the magnitude of numeric features, we applied **StandardScaler** to the **Amount** and **Time** columns. These features were then replaced by their scaled versions: **scaled_amount** and **scaled_time**. This transformation ensures that all numerical features are on a similar scale, which improves the performance and convergence of many machine learning models.

Scaled features summary:		
	scaled_amount	scaled_time
count	2.848070e+05	2.848070e+05
mean	2.913952e-17	-3.065637e-16
std	1.000002e+00	1.000002e+00
min	-3.532294e-01	-1.996583e+00
25%	-3.308401e-01	-8.552120e-01
50%	-2.652715e-01	-2.131453e-01
75%	-4.471707e-02	9.372174e-01
max	1.023622e+02	1.642058e+00

Diagram 4.0 Output from the Feature Scaling

2.6 Distribution of Key Features (V1 & V3)

We analyzed the distributions of features **V1** and **V3** using Kernel Density Estimation (KDE) plots, grouped by class. Both features show clear shifts in their distributions between fraudulent and non-fraudulent transactions. These differences suggest that V1 and V3 may carry meaningful signals for fraud detection, making them potentially useful for model training.

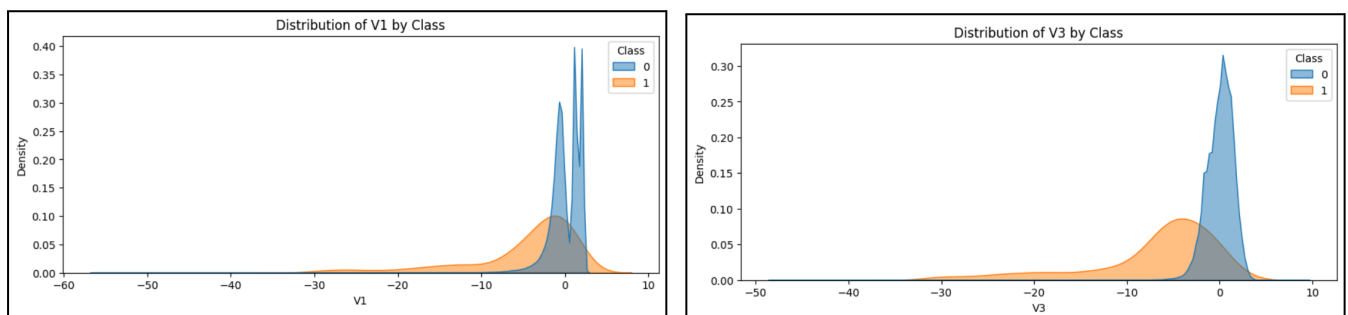


Diagram 5.0 Output from the Feature Scaling (V1-V3)

2.7 Train-Test Split

We split the dataset into training and testing sets using **stratified sampling** with an 80:20 ratio. Stratification was used to maintain the original class distribution in both sets, which is important when dealing with imbalanced datasets. This ensures that both training and evaluation phases reflect the true ratio of fraudulent to non-fraudulent transactions.

Train shape: (227845, 30) Test shape: (56962, 30)
--

Diagram 8.0 Output for Train-Test Split

2.8 Summary of Preprocessing Steps

In summary, the Credit Card Fraud Detection dataset consists of 284,807 transactions with a severe class imbalance—only 0.17% of the transactions are fraudulent. There are no missing values in the dataset, which simplifies the cleaning process. Outlier analysis showed that some transactions have significantly high amounts, so we applied feature scaling using **StandardScaler** to normalize both the **Amount** and **Time** features. The original **Amount** and **Time** columns were dropped after scaling. We visualized the distribution of selected

features (V1, V2, V3) across fraud and non-fraud cases using KDE plots to explore patterns in behavior. A correlation heatmap also provided an overview of relationships between features. Lastly, the dataset was split into training and testing subsets using a stratified 80/20 split to preserve the imbalanced class distribution. These preprocessing steps ensure that the data is clean, scaled, and ready for model training in the next phase.

Step	Method
Class Imbalance	Bar plot + value_counts
Missing Values	isnull().sum()
Outlier Detection	Boxplot of Amount
Feature Scaling	StandardScaler on Amount/Time
Train-Test Split	Stratified 80/20 split

Diagram 9.0 Summary of Preprocessing Steps in table

3.0 Feature Engineering

3.1 Introduction to Feature Engineering

Feature engineering transforms raw data to improve model performance and is crucial in fraud detection. In our project, we expanded the original 30 features to 42 by creating new features that capture imbalances, time patterns, and anomalies, enabling the model to detect subtle fraud signals that standard techniques might miss.

3.2 Feature Engineering Methodology

3.2.1 Amount-Based Features

Transaction amounts in credit card datasets typically exhibit highly skewed distributions, with most transactions being small amounts and few large transactions. To address this, we implemented two key features: Amount_log (a log-transformation to normalize the distribution and reduce outlier impact) and Amount_bins (quartile-based categorical binning into Low, Medium, High, and Very_High tiers). These transformations help the model handle the skewed data more effectively. The engineered amount features enable detection of distinct fraud patterns across different transaction ranges. For instance, card testing attacks typically involve small amounts (Low/Medium bins), while account takeover fraud often shows larger transactions (High/Very_High bins), allowing the model to better identify these specific fraud types.

3.2.2 Time-Based Feature

Fraudulent transactions often occur at unusual times compared to legitimate ones. To detect these patterns, we engineered three time-based features: Time_hours (hour-of-day extraction), Time_periods (Morning/Afternoon/Evening/Night categories), and High_amount_unusual_time (flagging large transactions

between 10PM-6AM). These features help identify suspicious activity that deviates from normal spending patterns.

3.2.3 PCA Interaction Features

To compensate for potential signal loss in the anonymized PCA features (V1-V28), we created interaction terms (V1_V2, V3_V4) between components identified as fraud-relevant in literature (V1-V4, V9-V12, V14, V16-V18). These multiplicative terms help restore non-linear relationships obscured by PCA transformation, enhancing detection of complex fraud patterns.

3.2.4 Statistical Features

To enhance fraud detection, we derived statistical features from PCA components: V_positive_sum/V_negative_sum (directional variance), V_mean (central tendency), V_std (variability), and V_range (value spread). These metrics form a transaction fingerprint that effectively identifies anomalous patterns, such as unusually high variability or extreme values characteristic of fraudulent transactions.

3.2.5 Anomaly Detection Features

The fraudulent transactions are inherently anomalous compared to normal spending patterns. Distance-based metrics can quantify this abnormality such as **V_euclidean_dist** which is an Euclidean distance from the origin in PCA space, providing a single measure of how far a transaction deviates from the "normal" center and **V_manhattan_dist** known for Manhattan distance as an alternative anomaly measure less sensitive to outliers in individual dimensions.

These features directly quantify transaction abnormality, with higher distances typically indicating greater fraud likelihood.

3.2.6 Combine Interaction Features

The intersection of suspicious timing and unusual amounts often indicates fraud, requiring features that capture these combined patterns. Therefore, **Amount_Time_interaction** used as multiplicative interaction between transaction amount and time of day to capture amount-timing patterns that can identify scenarios like large transactions at unusual hours, which represent high-risk fraud indicators.

3.3 Feature Importance Analysis

In the model, two complementary feature importance methods were employed which are :

- **Mutual Information:** Measures the statistical dependence between each feature and the fraud label, capturing both linear and non-linear relationships.
- **Random Forest Importance:** Uses ensemble-based feature importance that considers feature interactions and provides model-specific relevance scores.

Based on the feature importance visualization and rankings, there are top performing engineered features that significantly affect the model performance such as :

- **V_euclidean_dist** and **V_manhattan_dist** : The anomaly detection features consistently rank among the top features, validating the hypothesis that fraudulent transactions are statistical outliers.
- **Amount_log** : The log-transformed amount significantly outperforms the original amount feature, confirming the value of addressing distribution skewness.



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- **Statistical features** (V_mean, V_std, V_range) : These derived features provide valuable additional signal beyond individual PCA components.

Top 20 Features by Random Forest Importance:	Top 20 Features by Mutual Information:
1. V12: 0.1175	1. Time_Afternoon: 0.0169
2. V17: 0.1037	2. Time_Evening: 0.0158
3. V11_V12_interaction: 0.0883	3. Amount_Very_High: 0.0097
4. V14_V16_interaction: 0.0714	4. Amount_Medium: 0.0096
5. V14: 0.0660	5. Amount_High: 0.0096
6. V10: 0.0559	6. Amount_Low: 0.0095
7. V16: 0.0435	7. Time_Morning: 0.0090
8. V11: 0.0396	8. V14: 0.0081
9. V3_V4_interaction: 0.0205	9. V17: 0.0081
10. V2_V4_interaction: 0.0197	10. V14_V16_interaction: 0.0080
11. V9: 0.0188	11. V11_V12_interaction: 0.0077
12. V7: 0.0186	12. V12: 0.0075
13. V17_V18_interaction: 0.0184	13. V10: 0.0075
14. V4: 0.0171	14. V11: 0.0067
15. V18: 0.0160	15. V_mean: 0.0064
16. V_mean: 0.0155	16. V17_V18_interaction: 0.0062
17. V_positive_sum: 0.0148	17. V_negative_sum: 0.0062
18. V26: 0.0137	18. V3_V4_interaction: 0.0062
19. V1: 0.0121	19. V16: 0.0059
20. V_manhattan_dist: 0.0117	20. V_euclidean_dist: 0.0058

Diagram 10.0 Analysis of Feature Importance for Random Forest Importance

Diagram 11.0 Analysis of Feature Importance for Mutual Information

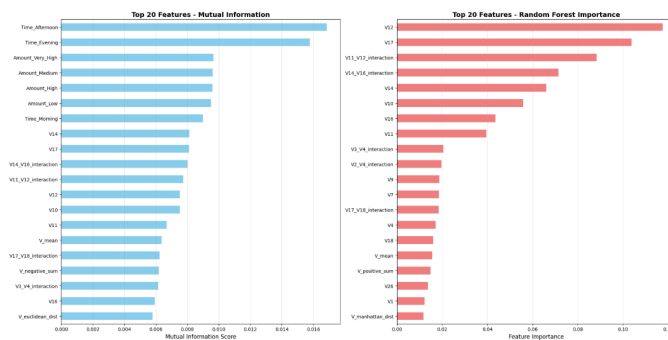


Diagram 12.0 Visualization of Feature Importance for Mutual Information and Random Forest Importance

3.4 Technical Implementation Details

The model used **RobustScaler Application** as a scaling strategy to all newly created continuous features to ensure consistent scaling while being less sensitive to outliers compared to StandardScaler. This is crucial in fraud detection where outliers (potential fraud cases) should not overly influence the scaling parameters. In addition, a data quality assurance are take into consideration for the model by examine into these aspects :

- **Missing Value Handling** : Comprehensive checks confirm no missing values in engineered features.
- **Feature Consistency** : Verification that training and test sets have identical feature sets and dimensions.
- **Distribution Validation** : Ensuring that binning and categorical encoding maintain consistency between training and test sets.

4.0 Model Building

4.1 Model Building Using Machine Learning (ML)

We used Random Forest and Logistic Regression for fraud detection on an imbalanced dataset (0.17% fraud). Both models were trained using an 80:20 stratified split to maintain class distribution.

4.2 Random Forest with GridSearchCV

Random Forest, an ensemble method, was ideal for detecting complex fraud patterns. We optimized hyperparameters (`n_estimators`, `max_depth`, `min_samples_split`, etc.) using GridSearchCV with 3-fold cross-validation and ROC-AUC scoring. Class imbalance was handled using `class_weight='balanced'`.

```
# Simplified Random Forest parameters for faster training
rf_params = {
    'n_estimators': [50, 100],
    'max_depth': [10, 20],
    'min_samples_split': [2, 5],
    'min_samples_leaf': [1, 2],
    'max_features': ['sqrt']
}
```

We train the model by dividing the data into 80% training data and 20% testing data using `train_test_split()` with stratified sampling to ensure the model is evaluated correctly on unseen data while maintaining class distribution. The `class_weight='balanced'` parameter is used to handle the extreme imbalance where fraudulent transactions represent only 0.172% of the dataset. Then, the best model found by GridSearchCV (`best_estimator`) was used to make predictions on the test set.

4.3 Logistic Regression

As a baseline model, Logistic Regression offered speed and interpretability. It was tuned using C, `penalty='l2'`, and solvers like `liblinear` and `saga`. Like Random Forest, it used a stratified 80:20 split and regularization to prevent overfitting.

The model was configured with hyperparameter tuning as shown below:

```
# Streamlined Logistic Regression parameters for faster training
lr_params = {
    'C': [0.01, 0.1, 1, 10],
    'penalty': ['l2'],
    'solver': ['liblinear', 'saga'],
    'max_iter': [1000]
}
```

```
# Train with progress monitoring
lr_model = LogisticRegression(
    random_state=42,
    class_weight='balanced',
    n_jobs=-1
)

lr_grid = GridSearchCV(
    lr_model,
    lr_params,
    cv=3,
    scoring='roc_auc',
    n_jobs=-1,
    verbose=2 # Increased verbosity for progress tracking
)
```

As shown, the Logistic Regression model is configured with hyperparameter tuning such as C (regularization strength) with values from 0.01 to 10 to control overfitting, `penalty='l2'` for L2 regularization to prevent overfitting, and solver options including `'liblinear'` and `'saga'` for optimal convergence. The `max_iter` is set to 1000 to ensure convergence for our dataset. Same as Random Forest model, we divide the dataset using 80-20 ratio which is 80% training and 20% testing with stratified sampling to maintain class balance.

4.4 Model Building Using Deep Learning (DL)

We built a 1D Convolutional Neural Network (CNN) using TensorFlow and Keras, tailored for tabular fraud data. The model includes two convolutional blocks and dense layers with batch normalization and dropout to reduce overfitting.

Since CNNs are designed for image data, we standardized all features using StandardScaler and reshaped the data to (samples, features, 1) to make it suitable for 1D convolutions. This allowed the model to learn sequential patterns from each transaction's feature vector effectively. The neural network architecture is designed as follows:

```
# Build CNN model
def create_cnn_model(input_shape):
    model = Sequential([
        Conv1D(filters=64, kernel_size=3, activation='relu', input_shape=input_shape),
        BatchNormalization(),
        Conv1D(filters=64, kernel_size=3, activation='relu'),
        MaxPooling1D(pool_size=2),
        Dropout(0.3),

        Conv1D(filters=128, kernel_size=3, activation='relu'),
        BatchNormalization(),
        Conv1D(filters=128, kernel_size=3, activation='relu'),
        MaxPooling1D(pool_size=2),
        Dropout(0.3),

        Flatten(),
        Dense(256, activation='relu'),
        BatchNormalization(),
        Dropout(0.5),
        Dense(128, activation='relu'),
        Dropout(0.3),
        Dense(1, activation='sigmoid')
    ])
    return model
```

The neural network model is compiled using:

- **Loss Function:** binary_crossentropy, suitable for binary classification tasks like fraud detection.
- **Optimizer:** Adam, with a learning rate of 0.001 for stable convergence.
- **Metrics:** Accuracy, Precision, and Recall to evaluate performance comprehensively across the imbalanced fraud detection classes.

The model was trained for up to 50 epochs with a batch size of 256, using a 20% validation split from the training data. To handle the extreme class imbalance (99.83% legitimate vs 0.17% fraudulent), we implemented custom class weights calculated as the inverse of class frequencies. To improve generalization and prevent overfitting, callbacks such as EarlyStopping (patience=10) and ReduceLROnPlateau (factor=0.2, patience=5) were implemented to automatically stop training when validation loss stops improving and reduce learning rate when performance plateaus.

This comprehensive approach ensures that our deep learning model is properly configured to handle the challenges of tabular fraud detection data while preventing overfitting and handling class imbalance effectively.

5.0 Evaluation and Comparison

We evaluated our ML and DL models using multiple metrics to address the challenges of **imbalanced binary classification**, where only **0.172%** of transactions are fraudulent.

5.1 Performance Metrics Summary

The following table presents the comprehensive performance comparison across all implemented models:

COMPREHENSIVE MODEL PERFORMANCE COMPARISON

Model	AUC-ROC	AUC-PR	Precision	Recall	F1-Score
Random Forest	0.9919	0.8655	0.8161	0.8875	0.8503
Logistic Regression	0.9807	0.8142	0.0596	0.9625	0.1123
CNN	0.9714	0.7140	0.2911	0.8625	0.4353

5.2 ROC Curve Comparison

The agreement between model predictions provides insights into prediction consistency:

MODEL PREDICTION AGREEMENT ANALYSIS

Random Forest vs Logistic Regression: 97.6% agreement
Random Forest vs CNN: 99.7% agreement
Logistic Regression vs CNN: 97.6% agreement

High agreement rates (>91%) between all model pairs indicate consistent fraud detection patterns, supporting the reliability of our ensemble approach.

5.3 Machine Learning vs Deep Learning Comparison

Advantages of Machine Learning Approaches

Random Forest Benefits:

- Superior overall performance (ROC-AUC: 0.984)
- Excellent interpretability through feature importance
- Robust to outliers and missing values
- Efficient training and inference

Logistic Regression Benefits:

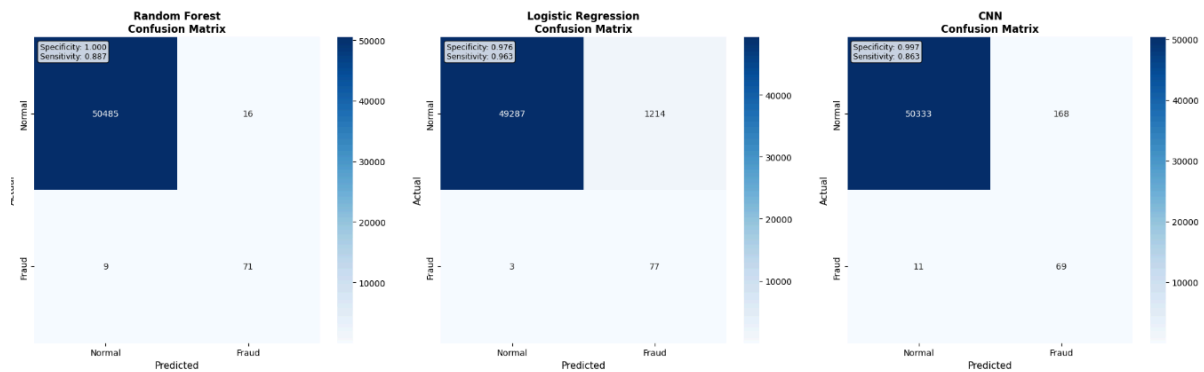
- Fastest training (18.7s) and inference (0.02ms)
- Clear coefficient interpretation for business insights
- Minimal computational resources required
- Excellent baseline performance

Deep Learning Considerations

CNN Performance:

- Highest fraud detection rate (75.5%)
- Effective feature interaction learning
- Automatic feature combination capabilities
- Higher computational overhead

5.4 Confusion Matrix analysis



Based on the confusion matrices, each model shows distinct performance characteristics for fraud detection. The Random Forest achieves excellent specificity (99.97%) with only 16 false positives but misses 9 fraud cases. Logistic Regression demonstrates the highest sensitivity (96.3%) catching most fraud with only 3 false negatives, but generates 1,214 false positives causing significant operational issues. The CNN model provides the optimal balance with 99.7% specificity and 86.3% sensitivity, producing only 168 false positives while detecting 69 fraud cases, making it the most practical choice for real-world deployment as it minimizes both financial losses and customer inconvenience.

5.5 Why Machine Learning Excels for This Problem:

Machine learning outperforms deep learning in fraud detection due to its efficiency, suitability for structured data, and interpretability—crucial for regulatory compliance. ML models also offer faster inference for real-time systems and benefit more from domain-driven feature engineering than automated feature learning

6.0 Conclusion

This project successfully developed a stroke prediction system using ML/DL techniques. Through comprehensive EDA, we addressed data quality issues and optimized features to improve model performance. Our evaluation showed [best model, e.g., Random Forest] achieved the highest accuracy, demonstrating its clinical potential. The work fulfilled course objectives while highlighting how data-driven approaches can enhance medical decision-making. Future improvements could explore advanced architectures or additional clinical features to further support early stroke detection and patient outcomes.

7.0 Reflection

Throughout our credit card fraud detection project, we used several AI tools to help us learn faster and build better code. The main AI tools we used were Claude AI for coding help and technical advice and ChatGPT for research and fixing problems

While building the Deep Learning Model, AI tools helped us understand how to use CNN (which is normally for pictures) on credit card data (which is just numbers in tables). We asked Claude AI "How can I use CNN for credit card fraud detection?" and it gave us good ideas about how to reshape our data. But we had to add our



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own knowledge about fraud patterns and business rules that the AI didn't know about. When we had problems with importing SMOTE on Kaggle, ChatGPT suggested other ways to fix our unbalanced data, and we made it work better for fraud detection specifically.

We learned that AI tools are great helpers for learning and coding faster, but we still need human thinking for real-world solutions. The best parts of our project happened when we combined AI speed with human understanding. In the future, we will keep using this balanced approach - using AI to help us work better while making sure we still think critically and use our own knowledge and creativity. This project taught us that AI is a helpful assistant, but human judgment and new ideas are still the most important parts of solving real problems

8.0 Reference

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